

Empirical Analysis and Mitigation of Vanishing Gradients in Deep Neural Networks

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Abstract—The vanishing gradient problem is a fundamental limitation in deep neural networks, particularly when using sigmoid activation functions. In this study, we empirically analyze gradient propagation in deep fully connected networks using sigmoid activations and compare it with a stabilized architecture using ReLU activations, improved initialization, and appropriate loss functions. Through controlled experiments, we demonstrate exponential gradient decay in sigmoid-based networks and show how modern architectural choices effectively mitigate this issue. Our results highlight the importance of activation functions, initialization strategies, and loss formulations in enabling stable deep learning.

I. INTRODUCTION

Deep neural networks rely on backpropagation to update parameters across multiple layers. However, as network depth increases, gradients may diminish exponentially, leading to ineffective learning in early layers. This phenomenon, known as the *vanishing gradient problem*, has historically hindered deep network training.

This study aims to:

- Empirically verify vanishing gradients in sigmoid-based networks
- Quantify gradient decay across layers
- Compare with modern techniques that mitigate the issue

II. METHODOLOGY

A. Experimental Setup

We designed two neural network pipelines:

1) Vanishing Gradient Setup:

- Activation: Sigmoid (all layers)
- Output: Sigmoid
- Loss: Binary Cross-Entropy (BCELoss)
- Optimizer: SGD (lr = 0.01)
- Initialization: Default

2) Stable Gradient Setup:

- Activation: ReLU
- Output: Linear (no sigmoid)
- Loss: BCEWithLogitsLoss
- Optimizer: Adam (lr = 0.001)
- Initialization: He (Kaiming) Initialization
- Input: Standardized

B. Model Architecture

Both models used identical depth:

$$\text{Input} \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 1$$

C. Gradient Measurement

Gradient norms were recorded using:

$$\|g\| = \sqrt{\sum g_i^2} \quad (1)$$

for each layer during backpropagation.

III. RESULTS

A. Gradient Analysis (Sigmoid Network)

Observed gradient magnitudes:

Layer	Gradient Norm
Layer 1	1.06×10^{-6}
Layer 2	1.93×10^{-6}
Layer 3	1.78×10^{-5}
Layer 4	1.95×10^{-4}
Layer 5	1.85×10^{-3}
Output	4.25×10^{-2}

TABLE I

GRADIENT DECAY IN SIGMOID NETWORK

B. Training Behavior

Loss evolution in sigmoid network:

$$0.73 \rightarrow 0.64 \rightarrow 0.72 \rightarrow 0.69 \rightarrow \dots \quad (2)$$

- Unstable convergence
- Slow improvement

C. Stable Network Observations

- Gradient magnitudes remained consistent across layers
- Loss decreased smoothly
- Faster convergence observed

IV. DISCUSSION

A. Why Vanishing Occurs

Sigmoid derivative:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \quad (3)$$

Maximum value:

$$\sigma'(x) \leq 0.25 \quad (4)$$

Thus, gradient propagation becomes:

$$g \propto (0.25)^L \quad (5)$$

where L is depth, leading to exponential decay.

B. Impact on Learning

- Early layers receive negligible updates
- Later layers dominate learning
- Network behaves like shallow model

C. Mitigation Strategies

- ReLU activation prevents saturation
- He initialization maintains variance
- BCEWithLogitsLoss improves numerical stability
- Input normalization avoids saturation regions

V. COMPARATIVE SUMMARY

Feature	Sigmoid Model	Stable Model
Gradient Flow	Vanishing	Stable
Loss Behavior	Noisy	Smooth
Learning Speed	Slow	Fast
Layer Utilization	Partial	Full

TABLE II
COMPARISON OF TWO PIPELINES

VI. CONCLUSION

This study empirically demonstrates that:

- Sigmoid-based deep networks suffer from vanishing gradients
- Gradient decay is exponential with depth
- Proper architectural choices mitigate the issue

In conclusion, this work demonstrates that the effectiveness of deep neural networks is governed not by depth alone, but by the stability of gradient flow, which is critically determined by activation functions, initialization schemes, and loss design.